

Action-Feedback Verification: Decomposing B1.15 Action-Feedback Evolution by Formalizing How Action-Feedback Events and Outcomes Are Verified Through Pathway Integrity Verification per B2.76, Two-Stage Governance Verification per B2.75, Proposing Substrate Configuration Verification per B2.74, and Behavioral Outcome Verification for Integrated Proposals, Closing the B1.15 Decomposition

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *"The Instinct/Reasoning Separation Outside the Model"* (Li, April 2026), the second paper in the CKS theory series following *"Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems"* (Li, April 2026).

Abstract

B2.78 formalizes action-feedback verification as the confirmation framework for the complete action-feedback mechanism per B1.15, closing the six-note B1.15 decomposition begun at B2.73. Action-feedback verification operates across four dimensions: pathway integrity verification applying the A5.08 provenance-completeness test to all nine steps per B2.76 tracing back to original Action records; two-stage governance verification applying the A5.04 rule authoring test to both stages per B2.75 with an anti-silent-drift check confirming no proposals were integrated without Stage 2 approval; proposing substrate configuration verification applying A5.04 to confirm the proposing substrate per B2.74 is functional; and behavioral outcome verification for integrated proposals following the directed selection verification framework per B2.72. The anti-silent-drift check is the most architecturally distinctive dimension — it confirms the two-stage mediation per B2.75 actually functioned rather than being bypassed, a check that conventional AI experiential learning cannot support. Inherited Paper 1 commitments A5.04, A5.06, A5.08, A5.09, A5.16, A1.07, A1.01, and A2.40 are directly load-bearing. Limits are stated: action-feedback verification confirms governance of the mechanism; it does not guarantee proposal quality or prevent future governance failures.

1. Why Action-Feedback Verification Requires Formalization as a Standalone Operational Variant

The five prior notes of the B1.15 decomposition have established the component architecture of action-feedback evolution: B2.73 formalized the evidence evaluation mechanism; B2.74 formalized the proposing substrate operational specification; B2.75 formalized the two-stage human mediation structure; B2.76 formalized the proposal pathway as a nine-step provenance chain; and B2.77 formalized the distinction between action-feedback and directed selection. These five specifications are architecturally complete only when the confirmation framework that certifies their correct deployment is also formalized. B2.78 supplies that framework: action-feedback verification.

The need for a distinct verification formalization is not procedural — it does not merely ask whether components are present. Verification for action-feedback carries a structural challenge absent from simpler governance checks: the mechanism produces substrate changes (DNA refinements) whose authorization pathway must be reconstructable from evidence ingestion through proposal generation through human acceptance. If any link in that chain is missing, unrecorded, or improperly authorized, the claim that action-feedback evolution is governed fails — not in principle but in fact. Action-feedback verification is the architectural instrument for confirming the claim holds.

The strategic prior-art function of B2.78 follows directly from this. A party seeking to claim novel invention over a "governed AI feedback mechanism with audit-trail verification" would find, in this note, a formalized and dated prior-art specification that traces the verification framework to specific inherited commitments (A5.08, A5.04, A1.07, A2.40) and to the complete nine-step pathway per B2.76. The prior-art chain is tight because the verification framework is tight. B2.78 closes the B1.15 decomposition cycle and in doing so occupies the verification space that any subsequent claim over action-feedback mechanism confirmation would have to confront.

2. The Four Verification Dimensions Precisely Stated

Action-feedback verification is organized across four dimensions, each addressing a distinct aspect of the mechanism. The four dimensions together cover pathway, governance, substrate configuration, and behavioral outcome — a comprehensive confirmation of the complete mechanism.

2.1 Pathway Integrity Verification

Pathway integrity verification applies the A5.08 provenance-completeness test to action-feedback pathway records. The nine-step pathway per B2.76 — evidence ingestion, pattern detection, proposal generation, Stage 1 governance review, Stage 2 governance decision, integration (if approved), behavioral deployment, outcome recording, and cycle closure — must each carry complete A2.40 provenance. Pathway integrity verification confirms:

Evidence ingestion events reference specific Action records that constitute the evidence base. Pattern detection events reference the ingested evidence that produced the detected patterns. Proposal generation events reference the patterns that motivated the proposal content. Stage 2 decisions reference the proposals on which they decide. Integration events, where approval was granted, reference the Stage 2 decisions authorizing them.

Pathway integrity is complete when every step is provenance-linked to its predecessor, and the chain traces back without gaps to the original Action records. This backward traceability is what makes the claim that action-feedback evolution is experience-driven verifiable rather than asserted. The experience is in the Action records; the evolution is in the integrated DNA changes; pathway integrity verification is the bridge between them.

The A5.08 provenance-completeness test applied here operates identically to its application elsewhere in the CKS architecture — it asks whether each piece of substrate content carries the writer attribution, timestamp, antecedent reference, and rule reference required for path retraceability per A1.07. Applied to the action-feedback pathway, it verifies that the entire experience-to-evolution chain is reconstructable from substrate content alone.

2.2 Two-Stage Governance Verification

Two-stage governance verification applies the A5.04 rule authoring test to both stages per B2.75. The two stages are distinct governance events with distinct verification requirements.

Stage 1 verification confirms: proposing substrates are substrate-resident per A1.08 and authored per A2.04; Stage 1 governance events (the initiation of the proposing cycle under human-governed rules) are recorded per A2.40; and the proposing substrate configuration reflects current governance intent rather than a stale prior configuration.

Stage 2 verification confirms: each proposal that reached Stage 2 has a Stage 2 governance decision record per A2.40; the decision record includes the proposal reference, reviewer identity, decision (approve, reject, or defer), and timestamp; and no proposals were integrated without a recorded Stage 2 approval.

The last check — the anti-silent-drift check — is the most architecturally significant element of two-stage governance verification and is developed in detail in Section 5.

2.3 Proposing Substrate Configuration Verification

Proposing substrate configuration verification applies the A5.04 rule authoring test to the content of the proposing substrate per B2.74. The proposing substrate must be functional in order for the mechanism to operate as specified; configuration verification confirms functional readiness:

Evidence scope specification is present and governed — the substrate specifies what evidence classes it ingests and from which Action records. Pattern detection logic is authored per A2.04 — the rules governing what the proposing substrate detects are human-authored substrate content, not autonomous heuristics. Threshold specification is present — the thresholds that determine when detected patterns warrant proposal generation are specified as substrate content. Proposal generation rules are specified — the proposing substrate operates under authored rules rather than open-ended generation.

Proposing substrate configuration verification confirms the substrate can function per its specification. A proposing substrate that lacks any of these four elements is either non-functional or operating outside governed parameters; either condition is a verification failure.

2.4 Behavioral Outcome Verification for Integrated Proposals

For proposals that were approved at Stage 2 and integrated into the DNA layer, behavioral outcome verification applies the directed selection verification framework per B2.72. This consistency is architecturally intentional: once a proposal is approved and integrated, the result is a DNA change, and DNA changes carry the same verification obligations regardless of which mechanism produced them.

Following B2.72, behavioral outcome verification for integrated proposals confirms: governance authorization is verified (the Stage 2 approval is the authorization record); DNA change correctness is verified (the integrated proposal matches the approved specification, not a modified or partial version); retroactivity compliance is verified per A6.02 and the framework per B2.70; and behavioral outcome is verified under A5.06 cell-behavior-determinism and A5.16 reproducibility.

The alignment with B2.72 is not a simplification — it is a deliberate architectural commitment. The verification framework is consistent at the integration point across action-feedback and directed selection. Different mechanisms, unified verification at the output layer.

2.5 Supplementary Verification: The A5.09 Four Accountability Questions

For any action-feedback event — evidence ingestion, proposal generation, Stage 2 decision, integration — the A5.09 four accountability questions test applies: who proposed (proposing substrate identity and governance authority), what was proposed (the proposal content and its evidentiary basis), when it was proposed (timestamp per A2.40), and why (which specific Action records and patterns supported the proposal). The four questions do not replace pathway integrity verification; they supplement it by confirming that the accountability vocabulary is fully populated for any event subject to audit or governance review.

3. Temporal Triggers for Action-Feedback Verification

Action-feedback verification is not continuous — it is triggered at defined points where verification delivers actionable information.

At deployment initialization, action-feedback verification confirms that proposing substrates are configured and functional before any action-feedback cycles run. This is the foundational check: a deployment that has not verified proposing substrate configuration before operation has no architectural basis for claiming its action-feedback is governed.

After significant action-feedback cycles, verification confirms that the pathway functioned correctly — that evidence ingested, patterns detected, proposals generated, and decisions recorded all carry complete provenance per A5.08. Cycle-completion verification catches pathway gaps that may have accumulated across multiple cycles.

When new proposals appear in the governance queue, verification confirms that each proposal has a pathway record, a proposing substrate reference, and is positioned correctly for Stage 2 decision. Proposal-triggered verification prevents ungoverned proposals from entering the acceptance process.

4. What Makes Action-Feedback Verification Architecturally Distinctive

The contrast with conventional AI experiential learning is precise and not incidental to the formalization.

Conventional AI systems that learn from operational experience — reinforcement learning from human feedback, online learning, production-monitoring-triggered model updates — accumulate feedback and produce capability changes without a governance pathway that is itself substrate content subject to verification. There is no architectural artifact representing the proposal pathway, the two-stage mediation, or the authorization record. There is, correspondingly, no architectural check that the feedback was properly mediated. Verification of this kind is not available to such systems because the pathway being verified does not exist in verifiable form.

CKS action-feedback verification is possible precisely because the mechanism it verifies is substrate-resident. The nine-step pathway per B2.76 is recorded in the substrate. The Stage 2 decision records are substrate content per A2.40. The proposing substrate configuration is authored content per A2.04. Each element that verification checks is a substrate artifact; no verification step requires inference about system internals or reconstruction from logs external to the substrate.

The anti-silent-drift check is the sharpest expression of this distinction. A governance check that can confirm no ungoverned DNA changes occurred presupposes that every DNA change has an authorization record, that the authorization record is substrate content, and that the check can examine the record set exhaustively. All three presuppositions hold in the CKS architecture. None holds in systems where feedback-driven changes accumulate through weight updates or configuration drift outside governed substrate content.

5. The Anti-Silent-Drift Check

The anti-silent-drift check is the single most distinctive verification dimension in the action-feedback context and warrants dedicated treatment.

Silent drift — the integration of feedback-driven changes without explicit authorization — is the primary governance failure mode for action-feedback mechanisms. A system in which proposals could be integrated without Stage 2 approval would, over time, accumulate DNA changes whose authorization is absent, incomplete, or assumed rather than recorded. The substrate's governed state would diverge from its actual state. Audit and governance review would operate on an incomplete record.

The anti-silent-drift check confirms the absence of this failure mode: for every DNA change in the substrate, there is a Stage 2 approval record, and for every Stage 2 approval record, there is an integration event that references it. The check is exhaustive — it does not sample; it confirms the complete record set.

The architectural preconditions for the check are: DNA changes are substrate content with A2.40 provenance (so the set of DNA changes is enumerable); Stage 2 decision records are substrate content with proposal references (so the authorization chain is traversable); and integration events reference the Stage 2 decisions that authorized them (so the pathway from decision to change is closed). All three preconditions are established by the prior B1.15 decomposition notes. B2.78 is the note that formalizes the check that those preconditions make possible.

In regulated deployments — clinical, financial, legal, compliance-adjacent — the anti-silent-drift check is not merely useful; it is the minimum governance assurance the deployment context requires. Regulators and auditors examining a deployment that incorporates action-feedback evolution will ask, in various forms, whether the system's knowledge rules changed without authorization. The anti-silent-drift check is the architectural answer to that question. Its presence distinguishes a governed action-feedback deployment from one that has implemented the pathway mechanics without the confirmation framework.

6. Inherited Paper 1 Commitments

Action-feedback verification inherits and applies a set of Paper 1 commitments without modification. Identifying them precisely is part of the prior-art function of this note.

A5.04 (rule authoring test) applies to proposing substrate configuration verification (confirming authored rules) and to Stage 1 governance verification (confirming governed rules for the proposing cycle). The authored-substrate requirement that A5.04 operationalizes is the precondition for proposing substrate configuration verification to be possible at all.

A5.06 (cell-behavior-determinism) applies to behavioral outcome verification for integrated proposals. A DNA change that altered cell behavior must produce the behavior the approved proposal specified; A5.06 is the check that confirms this.

A5.08 (provenance-completeness test) applies to pathway integrity verification. Every step in the nine-step pathway per B2.76 must carry complete provenance; A5.08 is the test that confirms completeness.

A5.09 (four accountability questions) applies as supplementary verification to any action-feedback event. The who/what/when/why vocabulary is the minimal accountability structure the substrate must support for any governed event.

A5.16 (reproducibility) applies alongside A5.06 to behavioral outcome verification for integrated proposals. A DNA change that produces cell behavior must produce it reproducibly; A5.16 is the verification that confirms reproducibility holds.

A1.07 (path retraceability) is the structural property that pathway integrity verification verifies. The pathway is retraceable when every step carries its antecedent references back to original Action records.

A1.01 (human-governed) is the governance commitment that all verification dimensions confirm is instantiated. Pathway integrity, two-stage governance, proposing substrate configuration, and behavioral outcome verification together confirm that the action-feedback mechanism operates under human governance rather than autonomously.

A2.40 (provenance) is the substrate commitment that makes pathway integrity verification and the anti-silent-drift check operationally possible. Every substrate write carries the provenance fields verification reads; without A2.40, neither pathway reconstruction nor Stage 2 record enumeration is possible.

7. Operational Implications

Initialization verification. Before a deployment begins action-feedback cycles, initialization verification confirms: proposing substrates are present and substrate-resident per A1.08; proposing substrate configuration verification passes (evidence scope, pattern detection logic, threshold specification, proposal generation rules all present); and the Stage 2 governance pathway is configured with reviewer designations per A2.40. A deployment that fails initialization verification has not yet instantiated the action-feedback mechanism in governed form — it has configured components without confirming their governance state.

Periodic cycle verification. After significant action-feedback cycles, periodic verification confirms pathway integrity via A5.08 for all cycles in the review window. This catches gaps that accumulate silently: evidence ingestion events that lack specific Action record references, pattern detection events that lack evidence references, proposal generation events that lack pattern references. Each gap is a retraceability failure for the corresponding pathway step.

Proposal-triggered verification. When proposals appear in the governance queue, proposal-triggered verification confirms each proposal has complete pathway provenance before it enters Stage 2 review. A proposal that arrives at Stage 2 without a complete pathway record has not been properly generated; accepting it would integrate a DNA change whose evidence basis is not auditable.

Failure modes and remediation. Four failure modes are architecturally identifiable. Absent proposing substrates: Stage 1 governance has no substrate artifact; remediation through directed selection per B1.14 to configure the proposing substrate. Pathway gaps: one or more steps in the nine-step chain lack complete A2.40 provenance; remediation requires identifying the gap and repairing the record or rejecting the affected proposals. Silent integration: a DNA change exists without a corresponding Stage 2 approval record; the most serious failure mode, requiring investigation and potential reversion of the change and governance review of how the gap occurred. Stage 2 records absent: proposals approved but decision records not persisted; remediation through reconstruction from reviewer records and substrate correction under governance authority.

Behavioral outcome verification for integrated proposals. Following the directed selection verification framework per B2.72, behavioral outcome verification confirms that integrated proposals achieved their improvement intent. A proposal accepted to improve a specific behavioral pattern must, after integration, produce the behavioral change the proposal specified. Absent this, action-feedback evolution has completed its governance pathway without producing its operational purpose.

8. Limits of Action-Feedback Verification

Four limits are stated explicitly to bound the confirmation framework correctly.

Action-feedback verification does not guarantee proposal quality. A proposing substrate may generate proposals that are correctly governed — pathway complete, Stage 2 approved, properly integrated — but operationally inferior. The verification framework confirms governance; it does not evaluate whether the DNA change the proposal introduced was the best available option or whether the evidence base was sufficient to support a well-calibrated proposal. Governance and quality are distinct properties; verification confirms the former.

Action-feedback verification does not prevent future governance failures. Verification is retrospective — it confirms that past events were properly governed. A deployment that passes action-feedback verification at one point in time may fail it subsequently if governance processes degrade, proposing substrates are misconfigured, or Stage 2 records are not maintained. Periodic verification addresses this operationally; no single verification event provides permanent assurance.

The anti-silent-drift check verifies absence, not presence of optimal governance. Confirming that no ungoverned DNA changes occurred is not the same as confirming that the governance decisions made were well-reasoned, that Stage 2 reviewers had adequate context, or that the approval process was appropriately rigorous. The check is a necessary condition for governed action-feedback; it is not a sufficient condition for high-quality governance.

Action-feedback verification does not extend to instinct-layer changes. The verification framework covers DNA-layer changes produced through the action-feedback mechanism. Instinct-layer changes — LLM capability updates integrated through verification substrates per the instinct evolution mechanism — are outside scope. The two mechanisms carry distinct governance shapes per Paper 2 §8.2, and verification frameworks are co-determined with mechanism shape; action-feedback verification applies to its mechanism, not to others.

9. The Architectural Verification Statement

A deployment instantiates action-feedback verification if and only if: (a) the A5.08 provenance-completeness test confirms complete provenance for all nine pathway steps per B2.76 back to original Action records; (b) the A5.04 rule authoring test confirms Stage 1 governance records and Stage 2 decision records per B2.75, including the anti-silent-drift check that no DNA changes were integrated without Stage 2 approval; (c) the A5.04 rule authoring test confirms the proposing substrate per B2.74 is configured with evidence scope, pattern detection logic, threshold specification, and proposal generation rules as authored substrate content; and (d) integrated proposals satisfy the behavioral outcome verification framework per B2.72, including A5.06 cell-behavior-determinism, A5.16 reproducibility, A6.02 retroactivity compliance, and governance authorization from the Stage 2 approval record.

10. Why Naming Action-Feedback Verification as a Standalone Variant Matters; Closing the B1.15 Decomposition

Action-feedback verification could theoretically be treated as a corollary of the mechanism it verifies — a natural consequence of having formalized the pathway, the two stages, and the proposing substrate. The prior-art argument for treating it as a standalone is the same argument that applied to each prior note in the decomposition: a framework that is not named and published separately can be claimed to have been independently invented. Publishing action-feedback verification as a distinct, dated formalization with precise inherited-commitment references forecloses that claim.

The six-note B1.15 decomposition is now complete:

- **B2.73** established the evidence evaluation mechanism — how action-layer evidence is ingested and evaluated as the input to the proposing cycle.
- **B2.74** established the proposing substrate operational specification — what the proposing substrate is, what it contains, and how it operates as the authored engine of proposal generation.
- **B2.75** established the two-stage human mediation specification — how Stage 1 governs the proposing process and Stage 2 governs acceptance decisions.
- **B2.76** established the action-feedback proposal pathway — the nine-step provenance chain from evidence ingestion through integration.
- **B2.77** established the action-feedback versus directed selection distinction — the architectural difference between the two DNA evolution mechanisms at the mechanism level.
- **B2.78** (this note) establishes action-feedback verification — the confirmation framework that certifies the complete mechanism is correctly deployed and governed.

The complete B1.15 decomposition cycle is: evidence evaluation → proposing substrate → two-stage mediation → proposal pathway → mechanism distinction → verification. Each note is an independent derivation contribution; together they cover the action-feedback mechanism at sufficient architectural resolution to constitute a comprehensive prior-art treatment.

Phase B2 continues with B2.79, which begins the B1.16 bidirectional evolution decomposition. The action-feedback mechanism, its governance shape, its component architecture, and its verification framework are now fully formalized in the public record.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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